Summary of this week

- Congratulations for completing the Kalman-filter Boot Camp!
- This week, you learned the steps of the Kalman filter.
- You learned how to prepare a system model for use by a Kalman filter.
- You learned how to implement that model in Octave code, and how to implement a Kalman filter in Octave code.
- You saw some examples that demonstrated the robustness of the Kalman filter to uncertain initial states, and how it outperforms open-loop predictors.
- You also saw some conditions that can cause a Kalman filter to fail.

Where to from here?

- Is it possible to overcome these failure conditions? Yes!
- But, to do so, you will need to learn how the KF is derived to see where the derivation assumptions are being violated.
- Then, we will be able to modify the derivation for special cases where we need to implement different assumptions.
- In Course 2, "Linear Kalman Filter Deep Dive,"
  - You will learn how to derive the KF equations.
  - You will learn how to overcome failure conditions.
  - You will learn additional applications of the KF and how to simplify the KF steps in some cases.
  - You will learn how to implement a target-tracking application with KF.

Credits

Credits for photos in this lesson

- "What's next?" stickie note on slide 2: Pixabay license (`https://pixabay.com/en/service/license/`), `https://pixabay.com/en/what-s-next-yellow-sticker-note-1462747/`